INCH - POUND
MIL-PRF-32635
09 October 2019

PERFORMANCE SPECIFICATION

CLOTH, FLAME RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for flame resistant (FR) cloth, for use in flame resistant clothing.

1.2 Classification. This specification covers the following five (5) types and classes.

1.2.1 Types.

Type I - Flame resistant, 65 percent rayon/25 percent para-aramid/10 percent nylon, ripstop
Type II - Flame resistant, 43 percent meta-aramid/30 percent nylon/27 percent cotton, twill
Type III - Flame resistant cloth, ripstop
Type IV - Flame resistant cloth, twill
Type V - Flame resistant antistatic cloth, twill

1.2.2 Classes. (see 6.10).

Class 1 - Operational Camouflage Pattern (OCP)
Class 2 - Coyote 498

Comments, suggestions, or questions on this document should be addressed to: Attn: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. N/A

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND SOLDIER CENTER

2-1-2592 - Operational Camouflage Pattern (OCP)

(Copies of drawings are available from the U.S. Army Combat Capabilities Development Command Soldier Center, ATTN: FCDD-SCP-WC, 10 General Greene Avenue, Natick, MA 01760-5019.)

FEDERAL TRADE COMMISSION

Rules and Regulations under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1, Gray Scale for Color Change
AATCC Evaluation Procedure 2, Gray Scale for Staining
AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale
AATCC Evaluation Procedure 9 (2018), Visual Assessment of Color Difference of Textiles
AATCC Test Method 8 - Colorfastness to Crocking: Crockmeter Method
AATCC Test Method 15 - Colorfastness to Perspiration

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AATCC Test Method 16.3 - Colorfastness to Light: Xenon Arc
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 20A - Fiber Analysis: Quantitative
AATCC Test Method 61 - Colorfastness to Laundering: Accelerated
AATCC Test Method 81 - pH of the Water-Extract from Wet Processed Textiles
AATCC Test Method 116 - Colorfastness to Crocking, Rotary Crockmeter Method
AATCC Test Method 135 - Dimensional Changes of Fabrics after Home Laundering
AATCC Test Method 143 - Appearance of Apparel and Other Textile End Products After Repeated Home Laundering

(Copies of these documents are available on line at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

ASTM INTERNATIONAL

ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1424 - Standard Test Method for Tearing Strength of Woven Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D3512/D3512M - Standard Test Method for Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Random Tumble Pilling Tester
ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M - Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM D5034 - Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Grab Test)
ASTM D5430 - Standard Test Method for Visually Inspecting and Grading Fabrics
ASTM D6413/D6413M - Standard Test Methods for Flame Resistance of Textiles Vertical Test)
ASTM F1930 - Standard Test Method for Evaluation of Flame Resistant Clothing for Protection Against Flash Fire Simulations Using an Instrumented Manikin

(Copies of these documents are available online at https://www.astm.org.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure
Principles and Methods of Toxicology, A Wallace Hayes (editor).

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(Copies of these documents are available online at https://www.crcpress.com.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 2112 - Standard on Flame-Resistant Garments for Protection of Industrial Personnel Against Flash Fire, 2007 edition

(Copies of this document are available online at https://www.nfpa.org.)

SDL ATLAS

Part Number 402985 – Slub/Knot Replica Set

(Fabric Defect Replica Set is available for purchase from SDL Atlas Customer Service 1-803-329-2110)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise specified, the finished cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Recycled, recovered, environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Basic material.

3.4.1 Fiber content.

3.4.1.1 Type I cloth. Type I cloth shall be made of warp and filling yarns using a fiber blend ratio of 65 percent flame resistant rayon, 25 percent para-aramid and 10 percent nylon.

3.4.1.2 Type II cloth. Type II cloth shall be made of warp and filling yarns using a fiber blend ratio of 43 percent meta-aramid, 30 percent nylon and 27 percent cotton.

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3.4.1.3 Type III and Type IV cloths. Types III and IV cloths shall be made of warp and filling yarns which result in a flame resistant cloth that meets the requirements of Table III and has received approval by the Contract Authority.

3.4.1.4 Type V cloth. Type V cloth shall be a blend of inherently flame resistant fibers, synthetic fibers, cellulosic fibers, and static-dissipative fibers which result in a flame- resistant cloth that meets the requirements of Table III and has received approval by the Contract Authority. Both the warp and filling yarns shall contain two (2) percent minimum static-dissipative fibers based upon the dry weight of the desized cloth. Testing shall be as specified in 4.5.

3.4.2 Weave.

3.4.2.1 Type I and Type III. The weave for Type I and Type III shall be plain weave with reinforcement ribs in both the warp and filling directions. The ribs for Type I and Type III should be formed by weaving two (2) warp ends as one (1) and two (2) picks as one (1) in the same shed, in an interval such that achieves the performance requirements of this specification.

3.4.2.2 Type II and Type IV. The weave for Types II and IV shall be 3 over 1 or 2 over 1 left-hand twill weave.

3.4.2.3 Type V. The weave for Type V shall be a 3 over 1 left-hand twill weave.

3.5 Color.

3.5.1 Visual shade matching (all Classes). The color and appearance of the dyed and camouflage printed cloth(s) shall match the standard sample when tested as specified in 4.5.

3.5.1.1 Class 1, Operational Camouflage Pattern (OCP). The finished cloth shall be either dyed to a ground shade matching Dark Cream 559 and then overprinted with the remaining six (6) colors Tan 525, Light Sage 560, Olive 527, Dark Green 528, Brown 529, and Bark Brown 561 areas of the camouflage pattern or when the ground shade is not dyed to Dark Cream 559 all seven (7) colors of the camouflage pattern shall be printed to match all seven (7) colors of the pattern.

3.5.1.2 Class 2, Coyote 498. The cloth shall be dyed to match the standard sample for Coyote 498 (see 6.4).

3.5.2 Colorfastness. The finished cloth for Classes 1, and 2 shall conform to the colorfastness requirements listed below in Table I and Table I.A, when tested in accordance with the methods specified in 4.5 and 4.6.2.

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TABLE I Colorfastness requirements, Class 1, all types.

Operational Camouflage Pattern (OCP)				
Colors Evaluation	Laundering (1 cycle) (color change and staining) (min.)	Light (40 AFU or 170 kJ /(m^2nm) @ 420 nm) (min.) 1/	Perspiration (acid & alkaline) (color change and staining) (min.)	Crocking Wet and Dry (min.)
All colors	-	-	3-4	-
Dk. Green 528, Brown 529, Bark Brown 561	2-3	3-4	--	Wet 2 Dry 3.5
Dark Cream 599, Tan 525, Light Sage 560, Olive 527	3	3	--	Wet 3 Dry 4

1/ AFU. AATCC Fading Units

TABLE I.A. Colorfastness requirements, Class 2, all types.

Coyote 498				
Colors Evaluation	Laundering (5 cycles) (color change and staining) (min.)	Light (40 AFU. or 170 kJ/(m^2nm) @ 420 nm) (min.)	Perspiration (acid & alkaline) (color change and staining) (min.)	Crocking Wet and Dry (min.)
Coyote 498	3-4	3	3-4	Wet 2.5 Dry 3

3.5.3 Labile sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The dyed cloth shall be "Free" of labile sulfur when tested as specified in 4.5 and 4.6.3.

3.6 Pattern execution (Class 1). The pattern on the printed finished cloth(s) shall match the standard sample in respect to design, colors and registration of the respective areas. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations and/or feathering as required for each color. Each pattern area shall show full coverage; skitteriness, feathering, haloing or trapping and off register exceeding that shown on the standard sample in any of the printed areas will not be acceptable. When the standard sample is not available for pattern execution, a pattern drawing will be provided (see 6.4) and the pattern on the finished cloth shall match that of the drawing 2-1-2592 when tested as specified in 4.5. The pattern repeat for OCP shall be as follows:

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Class 1 - Operational Camouflage pattern (OCP) - 25.255 (+1.25, -2.50) inches in the warp direction and 68-inches maximum camouflage printed area in the filling direction.

3.7 Spectral reflectance.

3.7.1 Class 1, Operational Camouflage Pattern (OCP). The spectral reflectance values shall conform to the requirements specified in Table II, when tested as specified in 4.6.4.

TABLE II. Spectral reflectance requirement. Class 1, all types (percent).

Operational Camouflage Pattern (OCP)						
Wavelength, Nanometers (nm)	Dark Cream 559 Tan 525		Light Sage 560 Olive 527, Brown 529		Dark Green 528 Bark Brown 561	
	Min.	Max.	Min.	Max.	Min.	Max.
600	22	44	12	30	3	11
620	24	45	12	30	3	12
640	24	45	12	32	4	13
660	25	45	12	32	4	14
680	28	45	14	34	4	17
700	28	48	14	36	6	23
720	30	52	16	39	6	23
740	32	55	18	41	10	25
760	36	56	20	43	14	30
780	38	57	22	45	18	35
800	40	57	22	45	21	40
820	44	58	24	46	24	42
840	46	59	26	47	26	43
860	48	60	28	48	28	45

3.7.2 Class 2, Coyote 498. The spectral reflectance of the color in the Coyote 498 cloth shall conform to the requirements specified in Table II.A, when tested as specified in 4.6.4.

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TABLE II.A. Spectral reflectance requirements, Class 2, all types (percent).

Coyote 498		
Wavelength, Nanometers (nm)	Min.	Max.
600	8	20
620	8	20
640	8	22
660	8	24
680	12	24
700	12	34
720	16	42
740	22	46
760	30	50
780	34	54
800	36	56
820	38	58
840	38	58
860	40	60

3.8 Physical requirements. The finished cloth(s) shall conform to the physical requirements listed below, in Table III, when tested in accordance with the test methods specified in 4.5.

TABLE III. Physical requirements.

Characteristic	Requirement				
	Type I	Type II	Type III	Type IV	Type V
Weight, oz./sq.yd. Min. Max.	 5.5 8.5	 5.5 8.5	 6.0 7.0	 5.7 7.2	 5.75 6.75
Breaking strength, pounds (min.) Warp Filling	 100 80	 100 80	 110 100	 110 100	 150 100

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TABLE III. Physical requirements. - Continued.

Characteristic	Requirement				
	Type I	Type II	Type III	Type IV	Type V
Tearing strength, pounds (min.)					
Dry					
Warp	4.0	4.0	12.0	12.0	10.0
Filling	4.0	4.0	10.0	10.0	12.0
Wet					
Warp	N/A	N/A	8.0	8.0	N/A
Filling	N/A	N/A	6.0	6.0	N/A
Dimensional stability, percent, (max.) After laundering (warp and filling) (1 cycle and 5 cycles)					
Individual Sample	4.0	5.5	4.0	5.5	5.0
Lot Average	3.5	5.0	3.5	5.0	4.0
Air permeability, cu.ft./min./sq.ft. (min.)	10.0	10.0	25.0	25.0	25.0
Flame resistance, initial:					
After flame, seconds (max.)	2.0	2.0	2.0	2.0	2.0
After glow, seconds (max.)	25.0	25.0	15.0	15.0	10.0
Char length, inches (max.)	4.5	4.5	4.5	4.5	5.0
Flame resistance, after laundering: (25, 50 and 100 cycles)					
After flame, seconds (max.)	2.0	2.0	2.0	2.0	2.0
After glow, seconds (max.)	25.0	25.0	15.0	15.0	10.0
Char length, inches (max.)	4.5	4.5	4.5	4.5	5.5
Thermal shrinkage resistance, percent (max.)					
Initial only	10	10	10	10	10
Heat Transfer Performance (HTP) (formerly Thermal Protective Performance (TPP))					
Spaced (min.)	N/A	N/A	10.0	10.0	10.0
Contact (min.)	N/A	N/A	7.0	7.0	7.0
Flash fire (Predicted Burn Injury), percent (max.)					
With head	37	37	35	35	35
Without head	30	30	28	28	28
Fabric appearance/smoothness, rating (min.)					
Initial	5	5	5	5	5
After laundering (15 cycles)	4	4	3	3	3

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TABLE III. Physical requirements. - Continued.

Characteristic	Requirement				
	Type I	Type II	Type III	Type IV	Type V
Pilling, 30 minutes (min.)	N/A	N/A	N/A	N/A	3.0
Electrostatic decay, seconds (max.) voltage level +/- 4000 volts (min.) Initial After laundering (5 cycles)	 N/A N/A	 N/A N/A	 N/A N/A	 N/A N/A	 0.5 0.5

3.9 pH. The pH of the water extract of the finished cloths shall be not less than 5.0 and not greater than 8.5 when tested as specified in 4.5.

3.10 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.7. The use of chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall be prohibited.

3.11 Face identification. The warp side shall be identified as the face side by stamping that side with the word “FACE” at each end of the roll.

3.12 Width. For Government procurements only, the width of the finished cloths shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage.

3.13 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put-up full width on a roll as specified in 5.1.

3.14 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.15 Workmanship. The finished cloths shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.3, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

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4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurements (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual Examination. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with MIL-STD-3064, Type I.

4.4.1 Roll identification and marking examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 End item testing. The cloth shall be tested for the characteristics listed in Table IV. The methods of testing as specified wherever applicable and as listed in Table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE IV. End item tests

Characteristic	Requirement Reference	Test
Fiber identification	3.4.1	AATCC-20 (see 6.8)
Fiber content	3.4.1	AATCC-20A (see 6.8)
Weave	3.4.2	Visual
Visual shade matching	3.5.1	4.6.1
Colorfastness to: Laundering Light Perspiration Crocking	Table I, I.A	4.6.2 AATCC- 61 3A 1/, 2/ AATCC-16.3, Opt 3 3/ AATCC-15 1/, 2/ AATCC-116, 4/ or AATCC-8 (Class 2 only), 4/
Presence of labile sulfur	3.5.3	4.6.3 5/(see 6.7)
Pattern execution	3.6	Visual and linear measurement using a Precision Steel Ruler
Spectral reflectance	Table II, II.A	4.6.4
Weight	Table III	ASTM D3776/D3776M (Method C)
Yarns per inch	Table III	ASTM D3775
Breaking strength	Table III	ASTM D5034
Tearing strength	Table III	ASTM D1424
Dimensional stability (after 1 and 5 laundering cycles)	Table III	AATCC-135, (3)(V)Aiii
Air permeability	Table III	ASTM D737
Flame resistance Initial, After laundering (25, 50 and 100 cycles)	Table III	ASTM D6413/D6413M AATCC-135, (3)(V)Aiii and ASTM D6413/ D6413M 6/
Thermal shrinkage resistance: Initial	Table III	NFPA 2112 (section 8.4) 5/
Heat transfer performance (formerly Thermal Protective Performance)	Table III	NFPA 2112 (section 8.2) 5/
Flash fire (Predicted Burn Injury)	Table III	ASTM F1930 (4 second exposure) 5/, 7/
Fabric appearance/smoothness: Initial After laundering (15 cycles)	Table III	AATCC-143 8/ AATCC-135, (3),(V),Aiii & AATCC-143 8/
Pilling (30 minutes)	Table III	ASTM D3512/D3512M
Electrostatic decay Initial After laundering (5 cycles)	Table III	4.6.5 AATCC-135 (3),(V),(Aiii) and 4.6.6
pH	3.9	AATCC-81
Toxicity	3.10	4.6

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1/ Only the stain on the nylon and cotton fibers of the color transfer cloth shall be evaluated..
2/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.
3/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change
4/ Rated using the AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale
5/ Required for First Article only (see 6.3)
6/ Tests after 50 and 100 cycles required for First Article only. Tests after 25 cycles required for lot testing.
7/ Material in applicable type and class shall be tested in accordance to applicable garment end item specification. For two piece uniforms, the coat or shirt shall be tucked into the waist or trouser. Manufacturer shall submit three (3) sets of garments to Product Manager Soldier Clothing and Individual Equipment (PdM-SCIE) for all Class 1, except for Class 2 where three (3) specimens will be sent to Product Manager Infantry Combat Equipment (PdM ICE) for testing.
8/ Rated using the AATCC Smoothness and Appearance Replicas.

4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the cloths shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option C (see 6.5), with sources simulating artificial daylight D75 illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200) K.

4.6.2 Colorfastness evaluation testing. When testing for colorfastness properties, each color shall be evaluated, whenever possible, separately and reported as such. In cases where the print pattern does not allow for the evaluation of each color separately, the test results should indicated which colors were evaluated together.

4.6.2.1 Operation Camouflage pattern (OCP) (Class 1). All colorfastness testing for Operation Camouflage Pattern shall be performed on the solid color area and not the tonal area.

4.6.3 Presence of labile sulfur test. In the determination of presence of labile sulfur in textile materials with lead acetate, two 1.50 (± 0.01) gram samples from each material submitted for evaluation shall be tested. Each of the two samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed in a stannous chloride solution that contains 100 grams of stannous chloride crystals ACS in 100 milliliters of hydrochloric acid ACS (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over the top of the test tube. The lead acetate solution contains 5.0 grams of lead acetate CP reagent grade and enough distilled water to make up a 100 milliliter solution; if the solution is not clear add a few drops (one at a time) of glacial acetic acid until the solution is clear. The test tube containing the textile sample, stannous chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

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Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail

4.6.4 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometer (nm) at 20nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates in the visible spectrum either CIE Source A or CIE Source D65. The specimen shall be viewed at an angle of no greater than 10º from normal, the specular component included. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The Operational Camouflage Pattern (OCP) (Class 1) and Coyote 498 (Class 2) shall be measured as a single layer backed with four (4) layers of the same shade cut from the standard. The spectral reflectance on OCP shall be measured on the mostly solid color area not the tonal area. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp and filling yarns. The diameter for standard aperture size used in the color measurement device shall be 1.0 to 1.25-inches for Class 2 and 0.3725-inches or larger for the Classes 1. The measured areas should be at least 6 inches away from the selvage. Photometric accuracy of the spectrophotometer shall be within one (1) percent and wavelength accuracy within two (2) nm. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.6.5 Electrostatic decay test (Type V only). The specimen shall be tested according to the method specified in 4.6.6.1 through 4.6.6.3 initially and after five (5) laundering cycles.

4.6.5.1 Specimen preparation. The specimen shall be a 3-inch by 6-inch rectangular piece of material. The warp or filling yarns shall run parallel to the 6-inch length of the specimen. The direction of test (warp or filling) shall be along the length of the specimen and each specimen shall be labeled accordingly in one (1) corner. Test specimens shall be cut so that no two (2) contain the same set of warp and filling yarns. Six (6) specimens, three (3) warp and three (3) filling, shall be tested, face and back sides inclusive.

4.6.5.2 Testing equipment. The test equipment shall consist of a meter for measuring relative humidity (RH) and temperature, a faraday cage, a source capable of outputting 5000 volts, and voltage meters to display the applied voltage and the voltage detected by the sensor. The faraday cage shall contain two (2) parallel electrodes on which specimens may be mounted and charged, and a sensor that will detect voltage on the specimen surface. The faraday cage and specimens to be tested shall be pre-conditioned at 10 (± 2) percent relative humidity overnight, and conditioned at 20 (± 2) percent relative humidity for a minimum of 24 hours at approximately

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75º (± 5)°F. The specimens shall be tested at 20 (± 2) percent relative humidity and 75º (± 5)°F. A bench top size chamber may be used as an option to achieve the conditions and to do the testing in, or any area capable of maintaining these conditions may be used. An air ionizing blower may be used while conditioning, although the blower must be turned off once testing begins. A static master may also be used to eliminate residual charges before testing.

4.6.5.3 Testing procedure. Allow the test equipment 1-hour of warm-up time. Adjust the voltage source for a charging voltage of 5000 volts by mounting a 3-inch by 5-inch aluminum plate across the electrodes in the faraday cage such that it is centered in front of the voltage sensor opening. Apply a charge of 5000 volts, and adjust/calibrate the voltage meter designated to indicate detected voltage on the sample equal to 5000 volts. Remove the aluminum plate. Mount a specimen tautly across the electrodes, centering it over the sensor opening. The surface of test specimen (back or front) shall face the sensor. The CUT-OFF must be set to 10 percent. With the high voltage off, depress the CHG button to measure initial charge on the test specimen. If it is not reading zero, turn CHG button off and use the static master or blower to attempt to eliminate the residual charge. With high voltage off, depress the CHG button again and record any remaining measurable residual charge. Turn off CHG button. Apply 5000 volts to the electrodes for a period of 20 seconds. To do this, depress the + button on the right (the reading on CHARGING VOLTAGE meter must read 5000 volts); then depress the CHG button. Wait 20-second; record the highest reading on the SAMPLE CHARGE meter. Then depress the TEST button and from the readout under DECAY TIME-SEC, record how long it took for the charge to decay from maximum voltage to 10 per cent of the maximum level. Depress the ZERO/STBY button after the test. The high voltage (5000 volts) shall be turned off and the specimen immediately grounded. If the DECAY TIME-SEC is less than 0.5 seconds, the specimen is acceptable; otherwise it is not. Also, the specimen must accept a charge of at least 4000 volts during the 20 second charging period to be acceptable.

NOTE. The reason for failure shall be noted. Recalibrate periodically, at the minimum between each set of six (6) specimens. Report the test method used and any alterations to the method shall be reported. Test conditions, including relative humidity, temperature and conditioning time shall be stated (see 6.6).

4.7 Toxicity test. When required, (see 6.2) an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If toxicity requirement (see 3.10) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the

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Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use in flame resistant, camouflage clothing.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type and class required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1, 4.2, 6.3)
e. Conformance inspection acceptance quality limits (AQLs)(see 4.3)
f. Camouflage pattern drawing, if required (see 2.2.2 and 6.4)
g. When toxicity testing is required (see 3.10, 4.7)
h. Width of cloth required (see 3.12)
i. Length required if other than specified (see 3.13)
j. Inspection conditions (see 4.3.1)
k. Packaging (see 5.1)
l. End item garment specification (s) and size (s) for Flash Fire testing (see Table IV, footnotes 5, 7 and 6.3)

6.3 First article. When a first article inspection is required (see 3.1.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to standard samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Visual shade matching. In 2019, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90o angle, relative to the plane of the specimens".

6.6 Electrostatic decay testing equipment known supplier: Electro-Tech Systems Inc., 115 E. Glenside Avenue, Glenside PA 19038.

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6.7 Replica set. The SDL Atlas Slub/Knot Replica Set is not available from their website at this time. Recommend calling customer service, 1-803-329-2110.

6.8 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.9 Superseding documents. This document supersedes GL-PD-07-12 Cloth, Flame Resistant (all revisions) and will be used in all future procurements.

6.10 Classes. Previous classes cited in GL-PD-07-12 Cloth, Flame Resistant have been changed to the following in this specification:

GL-PD-07-12	**MIL-PRF-**
Class 1- Universal Camouflage Pattern (UCP) (OCP)	Class 1-Operational Camouflage Pattern
Class 2-DELETED	Class 2-Coyote 498
Class 3- Operational Camouflage Pattern (OCP)	
Class 4-Coyote 498	

NOTE. Universal Camouflage Pattern (UCP) wear out date is September 30, 2019, therefore it was not included in the Military specification.

6.11 Subject term (key word) listing

Coyote 498
Dissipative Fiber
Operational Camouflage Pattern (OCP)

Custodians:
Army - GL
Air Force - 11

Preparing activity:

DLA-CT

(Project 8305-2019-010)

NOTE: The activities listed above were interested in this document as of date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST online database at https://assist.dla.mil.

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